SPICE simulation report: moon reading light (main board v0.1 · LED board v0.2)

Board versions: main board **v0.1**, LED board **v0.2** (`REV` in `hardware/*/design.py`, the KiCad title blocks and `fab/<board>/v<rev>/`). Report file: `spice-report_main-v0.1_led-v0.2`.

Generated 2026-11-08 by `python3 sim/run_sims.py` (ngspice-42, 45 runs over 10 circuits plus a model check). **These are predictions, not evidence**: every value nobody has measured is listed under *Assumptions* with the VERIFY item that replaces it.

Summary

19 pass, 5 marginal, 4 fail, 7 for information (35 checks against datasheet limits and the pass criteria in `docs/verification/VERIFY.md`). The FAIL rows are two issues: the USB-unplug dip (as built, plus the two R5 values tried as remedies) and the J3 hot-plug at a low battery.

What needs a decision:

1. **USB unplug dips VSYS by 0.64 V for about 21 ms** (VERIFY E12 allows 0.3 V). Q1's body diode carries the lamp until R5 has pulled VBUS down far enough to turn Q1's channel on. The MCU stays up (minimum 2.89 V against the 2.6 V BOD), but the LED-board current drops from 185 mA to about 130 mA for that time, which may show as a blink. E12's own remedy (R5 4.7k) only shortens it (10 ms; 2.5 ms with 1k): the depth is the body diode's drop and no R5 value changes it. See D1 below: accept for v0.1 (R5 1k if the dim shows); a reverse-blocking ideal diode in place of Q1 removes it in the next revision.
2. **Hot-plugging the LED-board cable resets the MCU at a low battery** (VSYS falls to 2.30 V at 3.49 V; 2.74 V on a full cell, just above the 2.6 V BOD). The cable is inside the gooseneck, so this is a bench situation, but VERIFY E13 asks for it. E13's remedy (47 µF on VSYS) brings the low-battery case to 2.72 V and raises the power-on inrush from 0.64 A to 1.68 A. Simplest: drop the hot-plug step from E13 (plug J3 with the switch off).
3. **On a weak USB source VSYS falls below E15's 3.9 V**: 3.71 V at 4.75 V USB through a 0.2 Ω cable with F1 at its 0.2 Ω maximum (about 3.80 V without the cable). Q1's body diode then sees 0.47 V forward and leaks 0.5 mA from the cell, too little to upset the charge termination. At 5.0 V VSYS is 4.00 V. E15 should name the source voltage (5.0 V nominal).

What the simulation confirms:

- Q4's soft turn-on works as designed: VLED ramps 10-90 % in 1.1 ms at 2.7 V/ms, and VSYS moves by 10 mV.
- Power-on through Q2 draws 0.64 A peak from a full cell (0.49 A at 3.49 V), with no overshoot on VSYS.
- Plugging USB in peaks at 5.77 V on VBUS and 4.78 V on VSYS (abs max 7 V / 6 V), even with a short, low-resistance cable.
- Load sharing (R18): with the lamp on, the cell still charges at 256 mA and tapers to the 31 mA termination after 5.1 h; USB draw peaks at 0.45 A.
- Battery life to the firmware's empty cut: 6.7 h at 4000 K full, 42 h at soft white level 10, 118 h in red night light (94-158 h for 12-7 mA of electronics). Close to the estimates in `docs/design/electronics.md` §3 (6.5 h, 45 h, 100-150 h).
- The whites stay regulated down to VSYS 3.10 V (typical dies) and 3.40 V (max-VF dies with the worst VSAT), below the 3.45 V low-battery cap. Green/blue need 3.57 V (typical) and 4.12 V (max die) at the board, matching the round-3 figures.

Findings and decisions to make

For main board v0.1 / LED board v0.2. Options give the simulated effect where it was simulated; "not simulated" marks an engineering estimate. Nothing in the design has been changed; each open item is Matt's call. *If no decision* says what happens by default.

#	FINDING	SEVERITY	STATUS	RECOMMENDATION
D1	USB unplug: VSYS dips 0.64 V for 21 ms	`FAIL` (E12)	open	v0.1: accept, re-word E12 (R5 1k if the dim shows). Next revision: reverse-blocking ideal diode
D2	J3 hot-plug at 3.5 V: VSYS 2.30 V, below BOD	`FAIL` (E13 step)	open: Matt confirms J3 is mated once at assembly	acceptable; mate J3 with SW1 off and drop the step from E13
D3	VSYS falls with the USB source voltage (3.71 V at 4.75 V, 3.36 V at 4.4 V)	`MARGINAL` (E15)	open	specify a 5 V / ≥ 1 A charger; E15 on 5.0 V; optional firmware tweak
D4	Max-VF white dies dim slightly below VSYS 3.40 V	`MARGINAL`	open (VERIFY E9)	accept; decide after E9
D5	I²C 419 ns at 150 pF	MARGINAL for fast mode only	**decided 2026-11-08**: keep 100 kHz	none
D6	Q4 charging current 36 mA vs 15-25 mA in the design note	INFO	**done 2026-11-08**: `electronics.md` §1 corrected	none

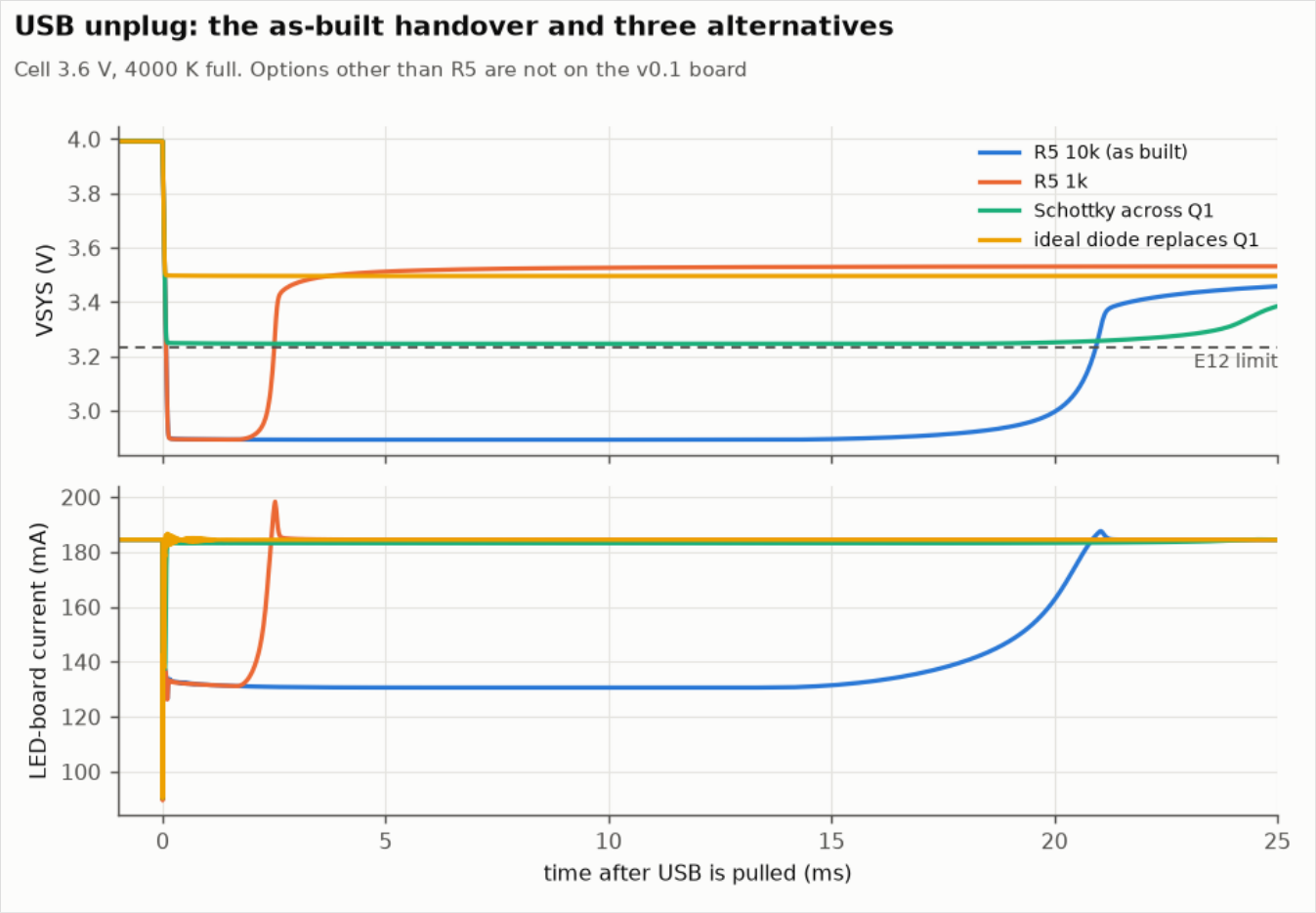
D1. USB unplug handover (VERIFY E12)

Finding. Pulling USB with the lamp at 4000 K full drops VSYS from 3.99 V to 2.89 V, 0.64 V below the 3.53 V it settles at on the battery, for 21 ms. The LED-board current falls from 185 mA to about 130 mA (−30 %) for 20 ms. The MCU stays above its 2.6 V BOD, so nothing resets; the risk is a visible dim at the moment of unplugging.

What happens, step by step (figure in section 5):

1. On USB, VSYS_RAW is fed through D3 at about VBUS − 0.85 V. Q1's gate sits on VBUS, so Q1 is off and the cell only charges.
2. At the unplug, the lamp drains C1 through D3 and VBUS falls quickly (4.85 → ~3.7 V in ~50 µs).
3. Once VSYS_RAW falls below the cell voltage minus ~0.65 V, Q1's **body diode** starts to carry the lamp. VSYS_RAW is then pinned at about 2.9 V and the diode drop sets the depth of the dip.
4. Q1's channel only turns on when VBUS < VSYS_RAW − |VGS(th)| ≈ 2.2 V. VBUS now sits on C1 (10 µF) and only R5 discharges it, so the time to get there is set by R5·C1. That is why R5 changes the length of the dip and not its depth.

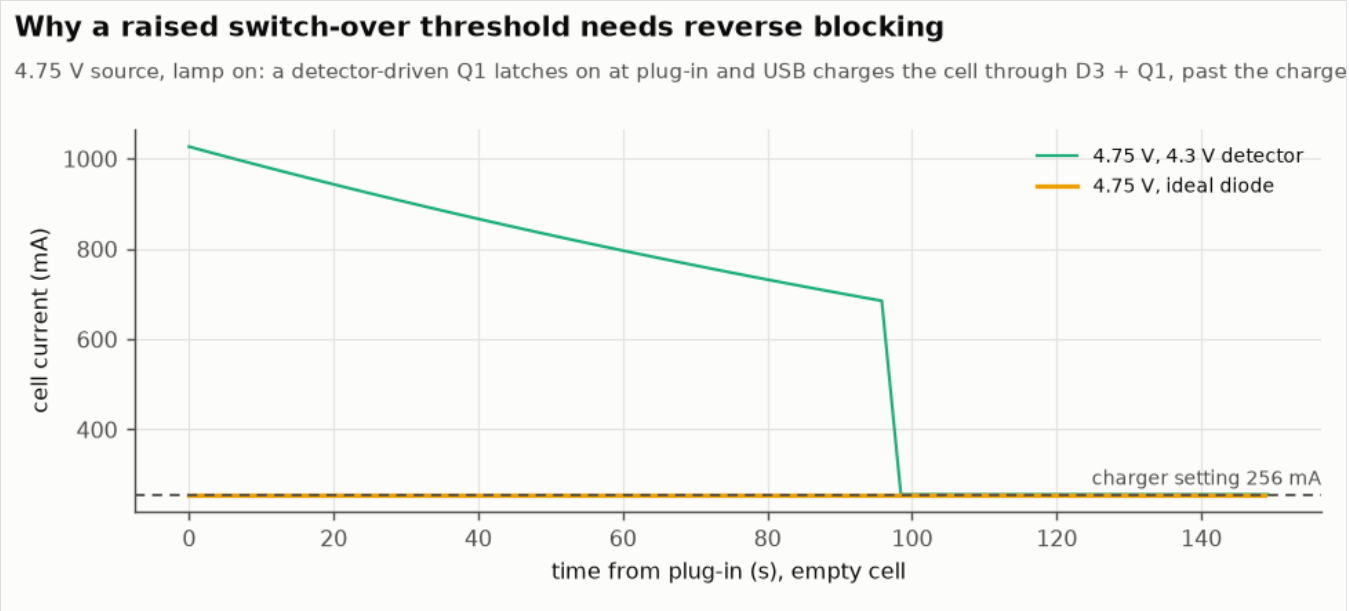
The root cause is the **switch-over threshold**: Q1 waits for VBUS to collapse to ~2.2 V, while USB has really gone as soon as VBUS drops below ~4 V. A fix either shortens step 4 or makes the switch-over happen at a higher VBUS.



VSYS and LED current after the unplug for the as-built handover and the alternatives that were simulated.

Options simulated (cell 3.6 V, lamp at 4000 K full; the weak-source column is the 4.75 V charge case):

OPTION	DIP DEPTH / TIME BEYOND 0.3 V	LED DIMS FOR	CHARGING ON A 4.75 V SOURCE	PARTS, COST, RISK
As built (R5 10k)	0.64 V / 20.8 ms	20.14 ms	as built (5.1 h)	-
R5 4.7k (E12's remedy)	0.64 V / 10.2 ms	9.87 ms	as built (5.1 h)	value change; +0.6 mA from USB while plugged
R5 1k	0.64 V / 2.4 ms	2.32 ms	as built (5.1 h)	value change; +4.5 mA (25 mW) from USB while plugged
Gate via a small Schottky, own 100k pull-down	0.64 V / 15.2 ms	14.56 ms	as built (5.1 h)	2 parts; **does not help**: VBUS itself stays up
1 A Schottky across Q1 (BAT → VSYS_RAW)	0.28 V / 0.0 ms	0.08 ms	terminates at 6.1 h	1 part (SOD-123); the battery shares the lamp whenever VSYS_RAW < BAT − 0.3 V
4.3 V detector drives Q1's gate	0.00 V / 0.0 ms	0.04 ms	**unsafe**: latches with Q1 on at plug-in; USB charges the cell through D3 + Q1 at up to 1.2 A, past the charger	detector IC + resistor
Reverse-blocking ideal diode replaces Q1, enabled below ~3.9 V VBUS	0.00 V / 0.0 ms	0.05 ms	terminates at 5.1 h	ideal-diode IC (LM66100-class, SC-70-6) + VBUS detection on its enable pin; layout change



The detector option on a weak source. Q1 is a bidirectional switch: once it is on while USB is present, D3 + Q1 form a path from USB straight into the cell. With a weak source this is a second stable state: the ~1 A it draws holds VBUS below the detector's release level, so the circuit stays latched (here for ~96 s, until the cell voltage rises). The current design avoids this only because Q1 waits for VBUS to collapse. A raised threshold therefore needs a switch that blocks reverse current.

Discussion.

- *R5* (B, C) is the only change possible on the v0.1 board without new parts. It cannot reduce the depth. With 1k the dim lasts 2.3 ms, probably too short to notice.
- *Schottky across Q1* brings the dip to 0.28 V, just inside E12. Its cost is the D3 interaction: on a weak source the cell feeds part of the lamp through the Schottky (charge ends at 6.1 h instead of 5.1 h). Its reverse leakage into the cell while on USB is µA-level at 25 °C and rises with temperature (not simulated). Note that the 2026-11-02 decision rejected a Schottky for D3 because of leakage with the switch off; across Q1 there is no bias with the switch off, so that concern does not apply here.
- *A detector on Q1's gate* removes the dip but is unsafe as shown above: rejected.
- *An ideal-diode IC* (e.g. TI LM66100: ~80 mΩ, reverse-current blocking, active-low enable, SC-70-6; check the datasheet and stock before choosing it) removes the dip completely (LED dim 0.05 ms) and keeps the weak-source charge unchanged (5.1 h). **The enable threshold matters**: VBUS falls to ~3.7 V within ~50 µs of an unplug, so a ~3.9 V threshold still catches it at once, while a higher one (4.35 V was tried) enables the ideal diode on a 4.4 V charger and hands the whole lamp to the cell: it then gains only ~70 mA net after 4 h and does not finish charging. It needs VBUS detection on its enable pin (a divider, or a voltage detector), a re-check of the standby drain (R17: its shutdown current adds to the ≤ 10 µA budget) and of the reversed-battery case, and a layout change. It is also a part that the hand-assembly rule R21 has to allow (SC-70 is fine by hand).
- *A charger with built-in power path* (MCP73871, BQ2407x) is the other next-revision route; those are QFN parts (R21) and were not simulated.

Recommendation. For v0.1, leave the hardware alone and re-word E12 to "no reset; note the dim". If the dim is visible at the bench, fit R5 = 1k. For the next main-board revision, replace Q1 with a reverse-blocking ideal diode enabled from VBUS.

If no decision. E12 fails at the bench against its 0.3 V limit; the lamp works.

D2. Hot-plugging the LED-board cable (VERIFY E13)

Finding. Plugging J3 while VLED is live sags VSYS to 2.30 V at a 3.49 V cell (below the 2.6 V BOD: expect a reset) and to 2.74 V on a full cell. The cable carries a 19 A pulse about 1.5 µs wide. Figure: section 3.

Matt's context (2026-11-08): J3 is mated once during assembly, then stays connected inside the casing where it cannot be reached.

Is that acceptable? Yes, with two conditions:

1. **Mate J3 with SW1 off** (VSYS is then 0 V and nothing is hot-plugged). Better still, mate it before the battery is plugged in. Then the case never happens in normal use.
2. **The connection must not come loose.** An intermittent J3 contact (for example from the gooseneck flexing) would repeat the hot-plug each time it reconnects: an MCU reset at a low battery, and the light coming back in its power-on mode. JST-SH has a friction lock only, so route the cable with strain relief and no tension on the plug. This is an annoyance, not a hazard: the 19 A pulse lasts ~1.5 µs ($I^2t \approx 3 \times 10^{-4}$ A^2s), far too short to heat the contacts.

Bulk capacitance on VSYS (47 µF) would only lift the low-battery case to 2.72 V and raises the power-on inrush from 0.64 A to 1.68 A, so it is not worth adding for a connector that is mated once.

Recommendation. Accept. Remove the hot-plug step from E13, and add "mate J3 with SW1 off" and a strain-relief note to the assembly handoff (05).

D3. VSYS on a weak USB source (VERIFY E15)

Finding. On USB, VSYS = VBUS − (cable + F1 drop) − D3's ~0.85 V, so it follows the charger's voltage and the cable. The firmware reads VSYS as "battery voltage", so a low VSYS on USB looks like a low battery to it. Charging with the lamp at 4000 K full (all cases through a 0.2 Ω cable, F1 at its 0.2 Ω maximum):

SOURCE AT THE PLUG	VSYS (MIN)	FIRMWARE SEES	CELL CURRENT THROUGH Q1'S BODY DIODE	CHARGE TIME
5.0 V	4.00 V	normal	0.0 mA	5.1 h
4.75 V (USB minimum at the port)	3.71 V	normal	0.5 mA	5.1 h
4.5 V	3.46 V	normal	94.5 mA	5.3 h
4.4 V (USB 2.0 bus-powered hub minimum)	3.36 V	**low battery** for 4.6 h (35 % cap, warning blinks)	157.5 mA	5.8 h

With the lamp off on a 4.4 V source VSYS is 3.65 V, below the firmware's 3.70 V resume level (VBAT_RESUME_MV): a lamp that went to empty-sleep would not wake until the charge current tapers.

Discussion.

- Nothing is unsafe at any of these voltages. The charger still regulates the cell, and when VSYS_RAW falls more than ~0.6 V below the cell, Q1's body diode lets the cell help carry the lamp (up to 158 mA at 4.4 V), which only stretches the charge.
- The user-visible effects are firmware ones: a false low-battery warning and a 35 % brightness cap on very weak sources, and a late wake from empty-sleep.
- A good USB-C charger delivers 5.0-5.2 V, and handoff 02 already asks for a ≥ 1 A charger. Weak sources are laptop ports through long thin cables, and bus-powered hubs.
- Firmware could avoid the false warning if it knew USB was present. The v0.1 board has no VBUS sense pin on the MCU (VBUS → divider → ADC would need a board change), so a firmware-only fix would have to infer USB from VSYS rising above the cell's possible range, which is not reliable.

- The D1 ideal-diode redesign (threshold ~3.9 V) does not raise VSYS on a weak source, since the lamp still runs through D3, but it stops the cell from feeding the lamp through Q1's body diode, so the charge at 4.4 V ends at 5.1 h instead of 5.8 h.

Options. (A) Specify the source: run E15 on a 5.0 V charger and state "5 V ≥ 1 A USB-C charger" in the user notes (no cost). (B) Lower VBAT_RESUME_MV to 3600 mV (E15's own remedy; firmware constant, costs a little hysteresis against bouncing out of empty-sleep). (C) Next revision: VBUS sense to an MCU pin, so the firmware can tell USB from a low battery.

Recommendation. A now; B if E15 shows the late wake on the chargers Matt uses; C with the D1 redesign.

D4. White-LED headroom with maximum-VF dies (VERIFY E9)

Finding. Each white channel needs VLED ≥ VF + VSAT for its full 25 mA. With typical dies that holds down to VSYS 3.10 V. With the highest-VF dies (2.92 V at 25 mA) **and** the worst LP5012 saturation voltage (0.45 V, assumed: not specified at 25 mA) it holds down to VSYS 3.40 V, just below the firmware's 3.45 V brightness cap. Figure: section 8.

How much it matters. Worst-case channel current from the headroom sweep: 24.2 mA (-3%) at the 3.45 V cap, 23.8 mA at 3.40 V, 22.5 mA (-10%) at the 3.30 V empty cut. This only happens in the last ~7% of the charge (at 4000 K full the cap arrives at 6.0 h, the cut at 6.7 h), when the firmware has already reduced brightness to 35 %, so a further few-percent dimming is hard to see. The cap lowers the average current but not the 25 mA peak per channel, so it does not restore the headroom by itself.

The one visible effect could be **colour**: if the 2700 K and 5000 K dies come from different VF bins, the higher-VF group dims first, and mixed whites drift slightly in the last few percent of the battery (not simulated; estimated as small next to the step from the 35 % cap).

Options. (A) Accept. (B) Raise VBAT_LOW_MV from 3450 to ~3500 mV, so the cap arrives before any channel can drop out (firmware constant; costs a little time at full brightness). (C) RIREF 3.65 kΩ (20 mA per channel, −20 % light), already listed under E9 for the green/blue dies. (D) Order a known low-VF bin (Bridgelux VF bins B-E); the stocked part is the full-distribution code, so this depends on availability.

Recommendation. A, and add the whites to E9: measure the received white dies' VF at 25 mA alongside G/B. If they sit near the top of the range, B is a one-line firmware change.

D5. I²C speed: decided

Rise times 168 / 280 / 419 ns for 60 / 100 / 150 pF are inside standard mode (1000 ns). **Decision (Matt, 2026-11-08): keep 100 kHz.** Moving to 400 kHz would need a measured bus capacitance below about 100 pF or stronger pull-ups.

D6. Documentation correction: done

`docs/design/electronics.md` §1 said Q4's soft turn-on charges the LED board with 15-25 mA. The simulation gives up to 36 mA, because the MLCCs start near their unbiased value. **Done (2026-11-08):** the note now gives the simulated 25-36 mA and the reason.

All checks

ID	AREA	CHECK	SIMULATED	LIMIT	VERDICT
P1-full	Power-on (SW1)	Cell inrush peak, cell 4.18 V	0.64 A	< ~3 A (typical DW01-class PCM over-current trip, PROVISIONAL); Q2 IDM 15 A	PASS

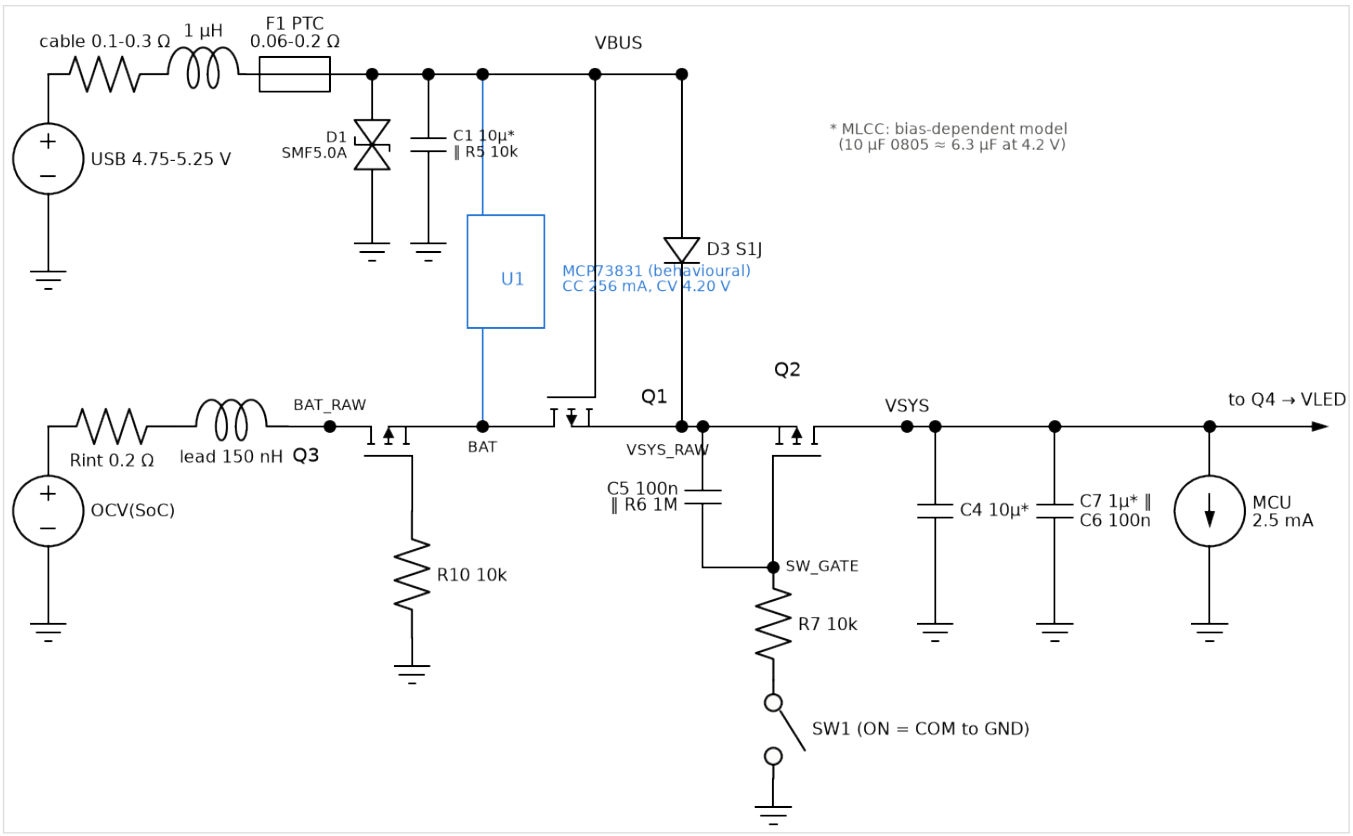
ID	AREA	CHECK	SIMULATED	LIMIT	VERDICT
P1-low	Power-on (SW1)	Cell inrush peak, cell 3.49 V	0.49 A	< ~3 A (typical DW01-class PCM over-current trip, PROVISIONAL); Q2 IDM 15 A	PASS
P1-full_bulk47	Power-on (SW1)	Cell inrush peak, cell 4.18 V, + 47 uF on VSYS	1.68 A	< ~3 A (typical DW01-class PCM over-current trip, PROVISIONAL); Q2 IDM 15 A	PASS
P2	Power-on (SW1)	VSYS rise 10-90 % and overshoot	76 us, overshoot 0 mV	monotonic, no overshoot	PASS
Q1	Q4 soft turn-on	VLED ramp at the LED board (10-90 %), cell 3.49 V	1.06 ms, 2.66 V/ms max	ramps over ~1-2 ms (E13); 2.5 V/ms design	PASS
Q2	Q4 soft turn-on	VSYS dip, worst of 4 corners	12 mV	<= 200 mV (E13)	PASS
Q3	Q4 soft turn-on	Peak LED-board charging current, worst corner	36 mA	design note says 15-25 mA	INFO
J1-full	J3 hot-plug	VSYS minimum, cell 4.18 V	2.74 V	> 2.6 V (ATtiny BOD level, fuses: bod = 2.6v)	MARGINAL
J1-low	J3 hot-plug	VSYS minimum, cell 3.49 V	2.30 V	> 2.6 V (ATtiny BOD level, fuses: bod = 2.6v)	FAIL
J1-low_bulk47	J3 hot-plug	VSYS minimum, cell 3.49 V, + 47 uF on VSYS	2.72 V	> 2.6 V (ATtiny BOD level, fuses: bod = 2.6v)	MARGINAL
J2	J3 hot-plug	Peak cable current / LED-board peak voltage	19.1 A / 4.18 V	LED board <= 6 V abs max (LP5012, MCP9800)	PASS
U1-r01	USB plug-in	VBUS peak, 5.25 V, cable 0.1 Ohm	5.77 V	< 7.0 V (MCP73831 abs max); TVS VBR 6.4 V min	PASS
U2-r01	USB plug-in	VSYS / LED-board peak, 5.25 V, cable 0.1 Ohm	4.78 / 4.76 V	< 6.0 V (ATtiny3216, LP5012 abs max)	PASS
U1-r03	USB plug-in	VBUS peak, 5.25 V, cable 0.3 Ohm	5.06 V	< 7.0 V (MCP73831 abs max); TVS VBR 6.4 V min	PASS
U2-r03	USB plug-in	VSYS / LED-board peak, 5.25 V, cable 0.3 Ohm	4.20 / 4.18 V	< 6.0 V (ATtiny3216, LP5012 abs max)	PASS
U3	USB plug-in	Attach inrush peak from the charger	10.1 A	USB 2.0: <= 10 uF at attach without inrush limiting	INFO
H1-r5_10k	USB unplug	VSYS dip below battery level, R5 10k (as built)	0.64 V for 21 ms (>0.3 V); min 2.89 V	<= 0.3 V, no reset (E12); BOD 2.6 V	FAIL
H1-r5_4k7	USB unplug	VSYS dip below battery level, R5 4.7k (E12 remedy)	0.64 V for 10 ms (>0.3 V); min 2.89 V	<= 0.3 V, no reset (E12); BOD 2.6 V	FAIL
H1-r5_1k	USB unplug	VSYS dip below battery level, R5 1k	0.64 V for 3 ms (>0.3 V); min 2.89 V	<= 0.3 V, no reset (E12); BOD 2.6 V	FAIL
H2	USB unplug	LED current during the handover (4000 K full)	185 mA -> 90 mA	no visible step (no spec)	INFO
C1	Charge + load share	Charger terminates with the lamp on (R18)	I_bat falls to 31 mA at 5.09 h; Q1 body diode max 0.19 V forward	termination reached; Q1 body diode < 0.4 V (not conducting)	PASS
C2	Charge + load share	Same, weak USB (4.75 V, PTC 0.2 Ohm cold max, cable 0.2 Ohm)	Q1 body diode 0.47 V forward, 0.46 mA from the cell into VSYS; terminates at 5.09 h	body-diode current < 3.1 mA (10 % of the 31 mA termination)	PASS

ID	AREA	CHECK	SIMULATED	LIMIT	VERDICT
C3	Charge + load share	Peak USB current, charging + 4000 K full	0.447 A	<= ~0.8 A (F1 hold at 60 C); E11 expects 0.5-0.6 A	PASS
C4	Charge + load share	VSYS on USB (minimum over the charge)	4.00 V (5.0 V USB); 3.71 V (4.75 V, PTC 0.2 Ohm) (both with a 0.2 Ohm cable)	>= 3.9 V (E15)	MARGINAL
C5	Charge + load share	MCP73831 dissipation, peak	0.46 W	<= 0.69 W design worst case (thermal regulation above)	PASS
R1-4000k	Runtime	Battery life, 4000 K full	6.7 h to the empty cut (4.45 Wh delivered; brightness cap after 6.0 h)	estimate 6.5 h (electronics.md 3)	INFO
R1-soft10	Runtime	Battery life, soft white level 10	42.0 h to the empty cut (4.50 Wh delivered; brightness cap after 40.1 h)	estimate 45 h (electronics.md 3)	INFO
R1-night	Runtime	Battery life, red night level 4	118.2 h to the empty cut (4.51 Wh delivered; brightness cap after 115.1 h)	estimate 100-150 h (electronics.md 3)	INFO
D1	LED headroom	Whites stay regulated down to the low-battery cut (8 on together)	95 % point at VSYS 3.10 V / cell OCV 3.17 V (typ); VSYS 3.40 V / OCV 3.47 V (max-VF dies, VSAT 0.45 V)	VSYS edge < 3.45 V (VBAT_LOW is measured on VSYS under load)	MARGINAL
D2	LED headroom	Green/blue regulation edge (VLED at the board)	typ die 3.57 V; max-VF die 4.12 V	design: ~3.65 V typ, ~4.05-4.15 V max die (round3 R3-E3)	INFO
W1-d50	PWM ripple	Ripple, all whites in phase, 50 % duty	VLED 82 mV p-p (min 3.65 V); VSYS 59 mV p-p	LP5012 VCC >= 2.7 V; VSYS > 2.6 V	PASS
W1-d89	PWM ripple	Ripple, all whites in phase, 88.75 % duty (4000 K full)	VLED 48 mV p-p (min 3.64 V); VSYS 30 mV p-p	LP5012 VCC >= 2.7 V; VSYS > 2.6 V	PASS
I1-c60	I2C	Rise time 30-70 %, 60 pF	168 ns	<= 1000 ns (standard mode, 100 kHz)	PASS
I1-c100	I2C	Rise time 30-70 %, 100 pF	280 ns	<= 1000 ns (standard mode, 100 kHz)	PASS
I1-c150	I2C	Rise time 30-70 %, 150 pF	419 ns	<= 1000 ns (standard mode, 100 kHz)	MARGINAL

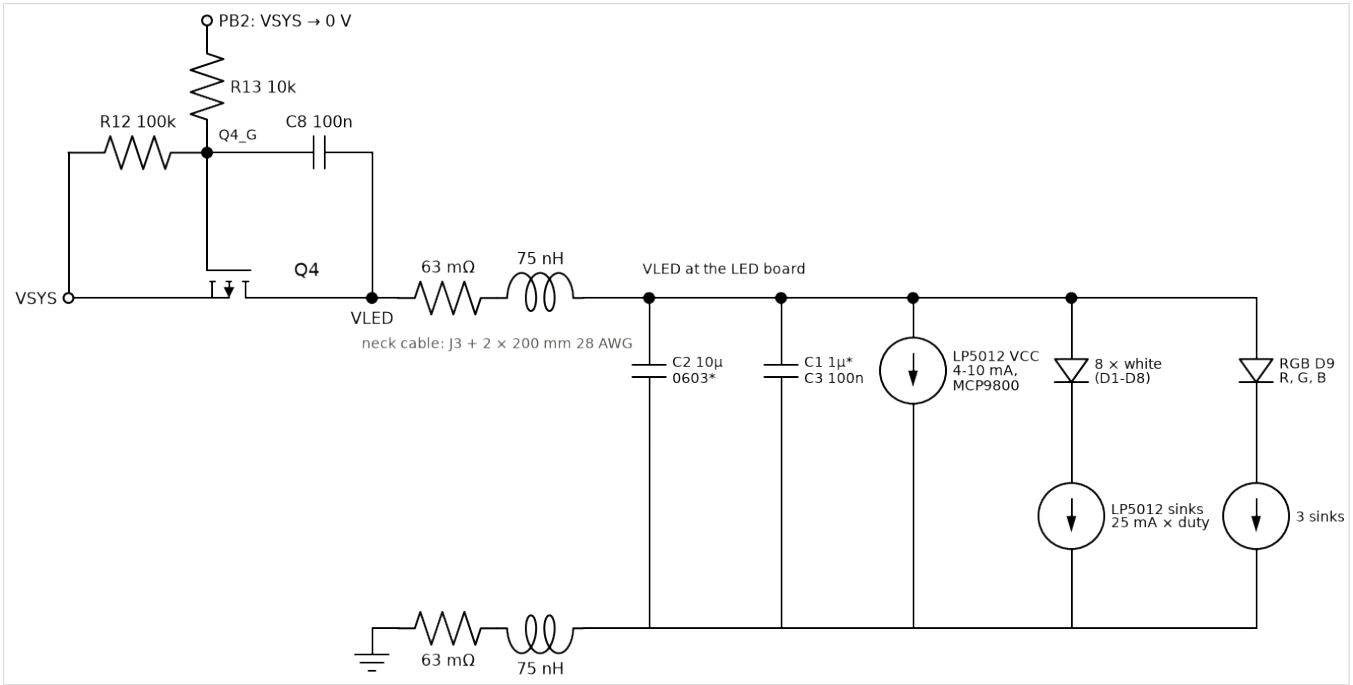
MARGINAL means inside the limit with less margin than this report's own rule of thumb (stated per check in `sim/run_sims.py`). INFO rows have no hard limit.

The circuits as simulated

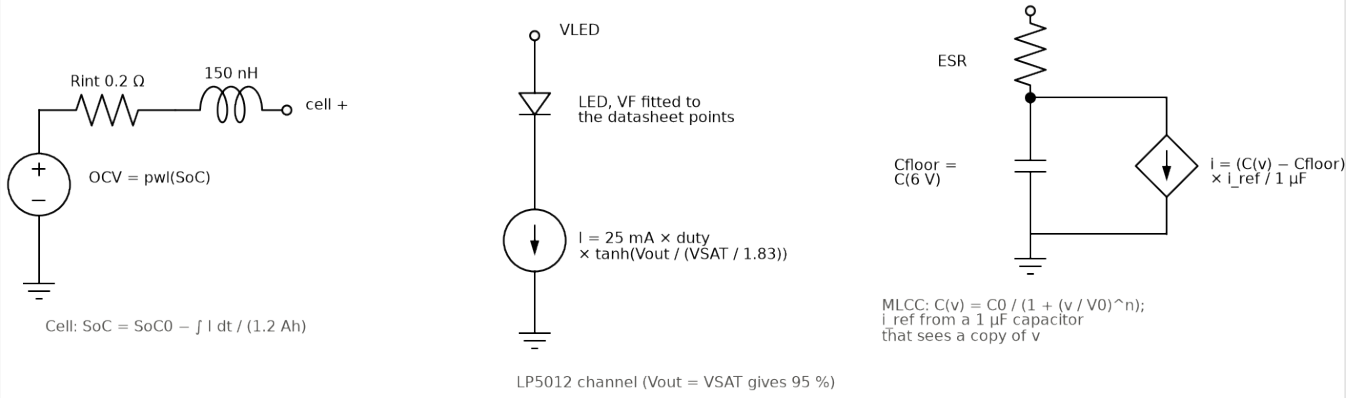
SPICE subcircuits in `sim/boards.inc` follow the reference designators of the two boards. Device models are in `sim/models.inc`; each is fitted to datasheet points (checked in the *Model check* table below).



Main-board power path: cell model, reverse-battery FET Q3, load-share FET Q1 (gate on VBUS), D3 from USB, the slide-switch FET Q2 with its R7/C5 soft start, and the VSYS loads. The charger is a behavioural CC/CV source.



Q4 with its Miller soft start (R13 into C8), the neck cable (both conductors), and the LED board: bulk and bypass MLCCs, LP5012 supply current, and the LED channels as LED + LP5012 sink.

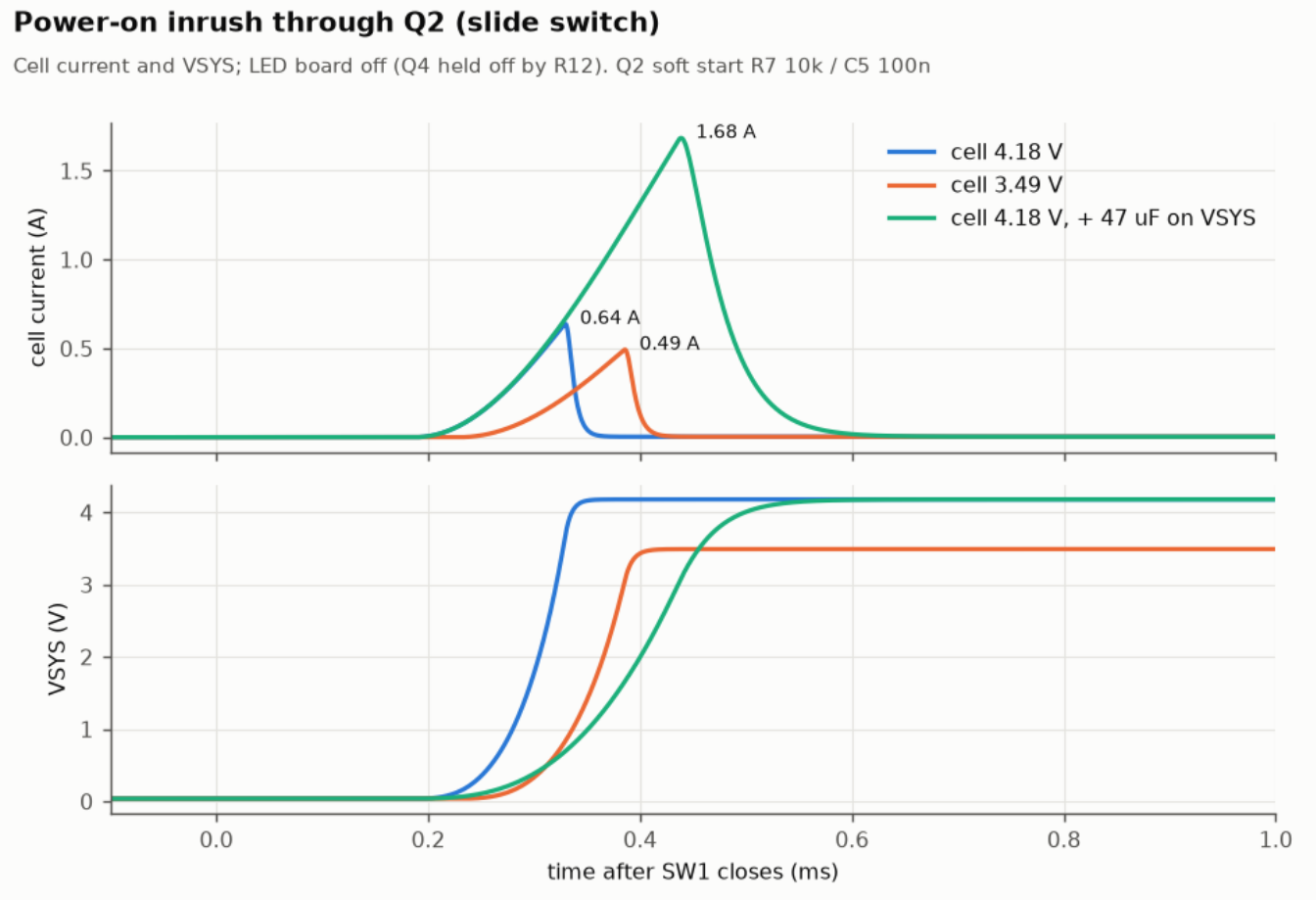


Behavioural models: the cell (open-circuit voltage from the state of charge), one LP5012 channel, and the bias-dependent MLCC.

1. Power-on inrush (slide switch, Q2)

SW1 pulls Q2's gate down through R7 10k; C5 (100 nF) sits gate-to-source, so the gate voltage falls with a 1 ms time constant and Q2 turns on hard once it passes its threshold. The VSYS capacitors (C4 10 µF, C7 1 µF, C6 100 nF; about 7 µF at 4 V) then charge through the cell, Q3, Q1 and Q2.

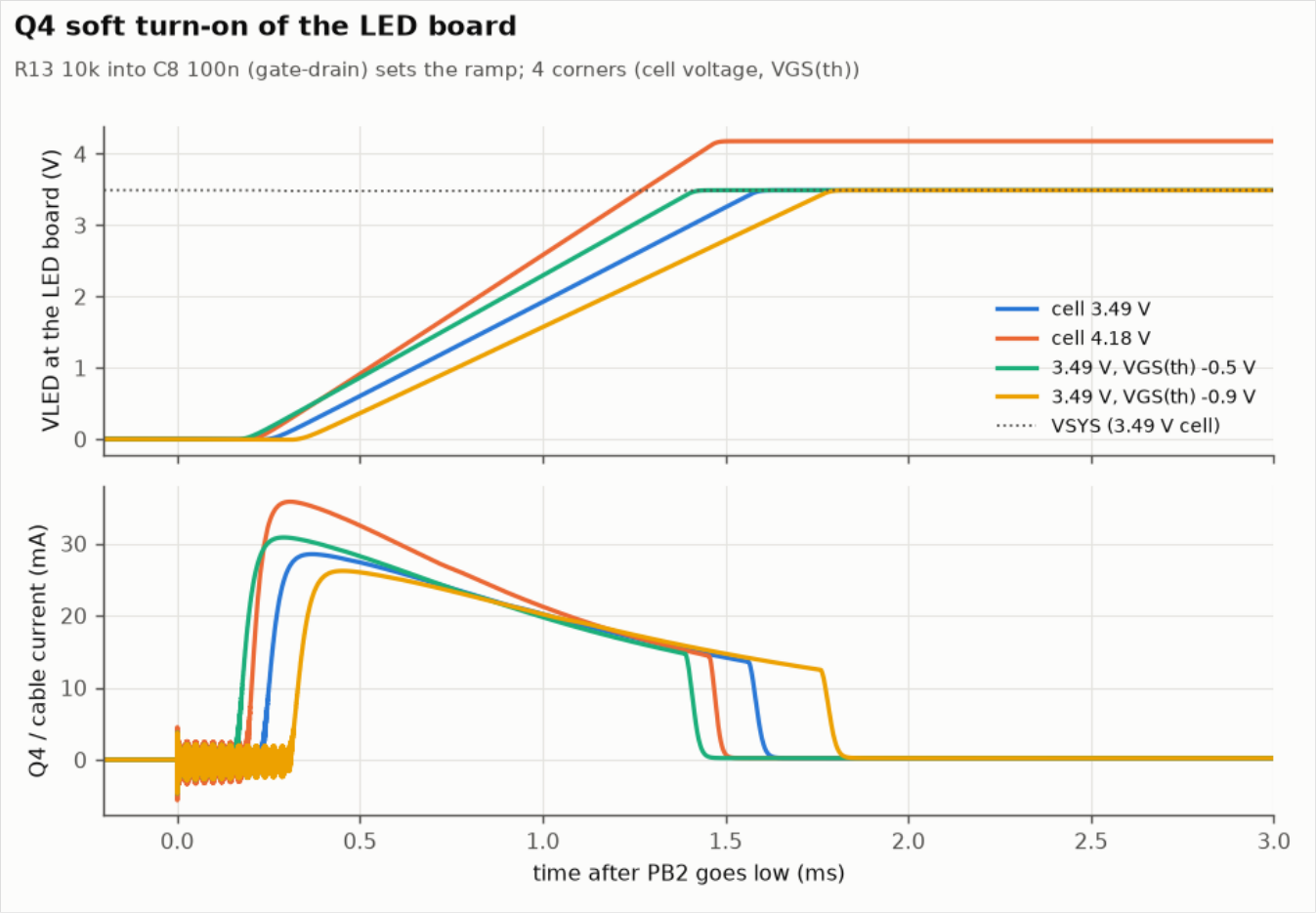
Peak cell current 0.64 A (full cell) and 0.49 A (3.49 V); VSYS rises 10-90 % in 76 µs with no overshoot. Round 1 had predicted 0.7-1.3 A into 14-21 µF; the LED board now sits behind Q4, so Q2 sees less capacitance. With 47 µF added on VSYS (the E13 remedy) the peak rises to 1.68 A, still well below the FET's 15 A pulse rating but closer to the cell PCM's over-current trip (typically ~3 A on DW01-class protectors; not documented for this cell).



2. Q4 soft turn-on of the LED board

PB2 drives Q4's gate through R13 10k with C8 100 nF from gate to drain: the Miller plateau holds the gate while C8 charges, so VLED rises at about (VSYS − |VGS(plateau)|) / (R13 · C8). The LED board is in standby (LEDs off), so it loads VLED with its MLCCs only. Four corners: cell 3.49 V / 4.18 V and Q4 threshold −0.5 / −0.9 V.

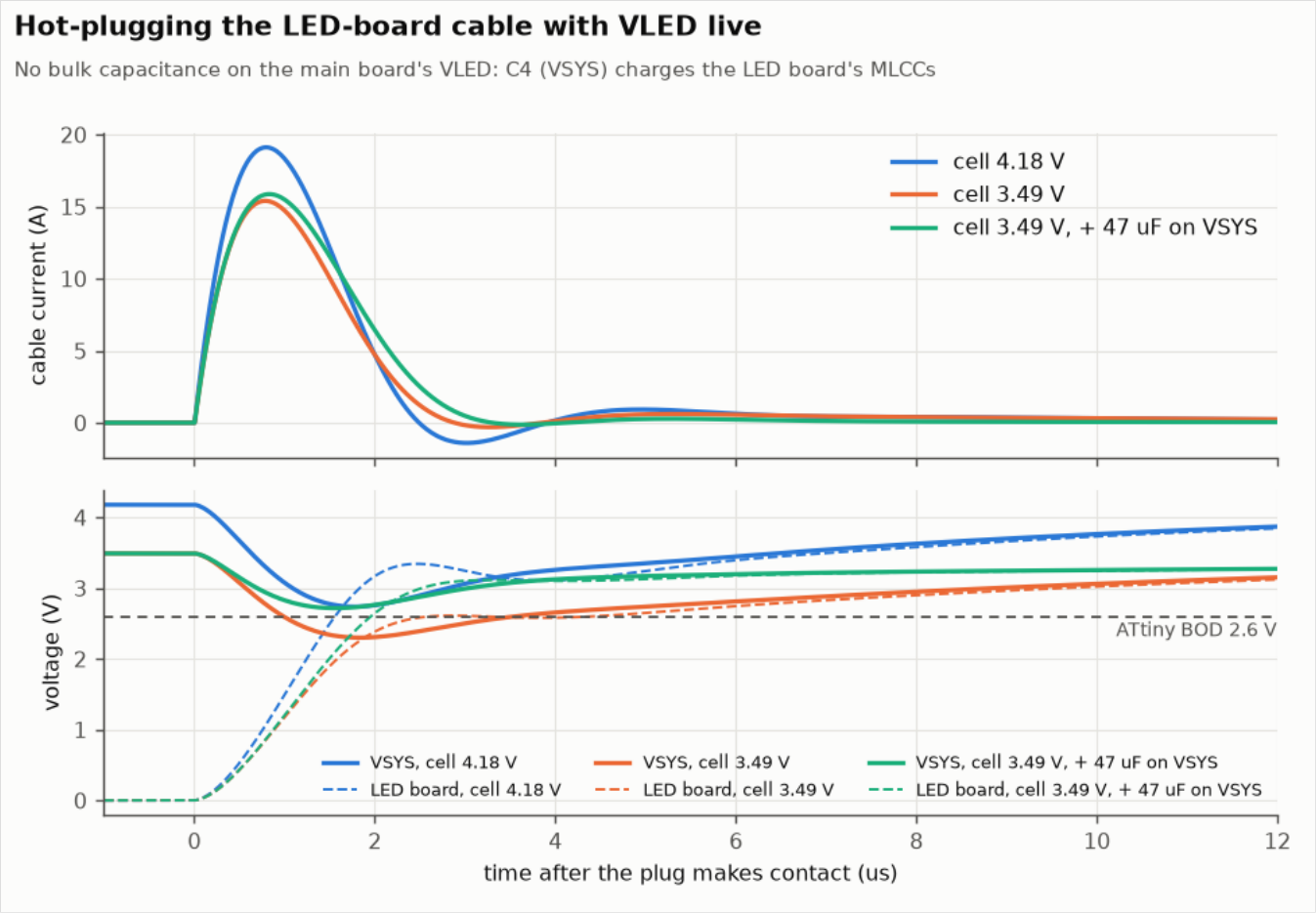
At 3.49 V (the E13 condition) VLED takes 1.06 ms from 10 to 90 %, peak 2.66 V/ms, and the charging current peaks at 29 mA (36 mA worst corner). VSYS moves by 10 mV. The current is higher than the design note's 15-25 mA because the MLCCs start at their unbiased value (about 11 µF at 0 V, falling to about 4.5 µF at 3.5 V); it is still harmless.



3. Hot-plugging the LED-board cable

With Q4 on, plugging the cable connects the LED board's empty MLCCs straight to VSYS. The main board has no bulk capacitance on VLED, so C4 shares its charge with them within about 2 µs; the cell, behind its resistance and lead inductance, brings VSYS back over the next ~50 µs.

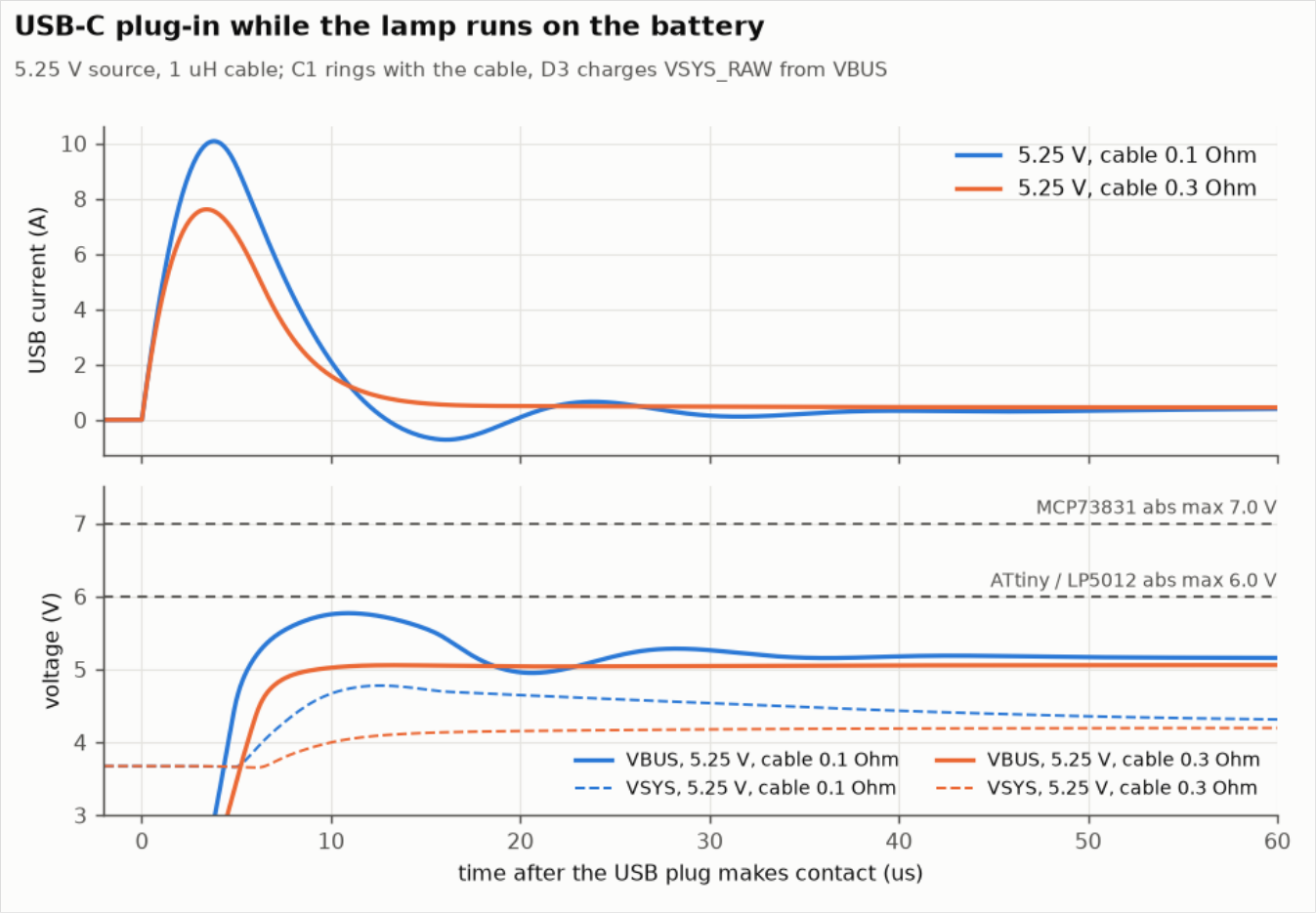
The cable current peaks at 19 A in a pulse about 1.5 µs wide. VSYS falls to 2.74 V on a full cell and to 2.30 V at 3.49 V, below the ATtiny's 2.6 V BOD level, so expect a reset. With 47 µF on VSYS the 3.49 V case bottoms at 2.72 V. The BOD's response time to a dip of a few microseconds is not modelled; a short dip may not trip it, so treat this as the worst case. The LED board never exceeds VSYS (no overshoot).



4. Plugging in USB while the lamp runs

A 5.25 V source (top of the USB tolerance) through a 1 µH cable is connected while the lamp runs at 4000 K from a 3.74 V cell. The cable rings with C1, and D3 also charges VSYS_RAW/VSYS up from the battery level.

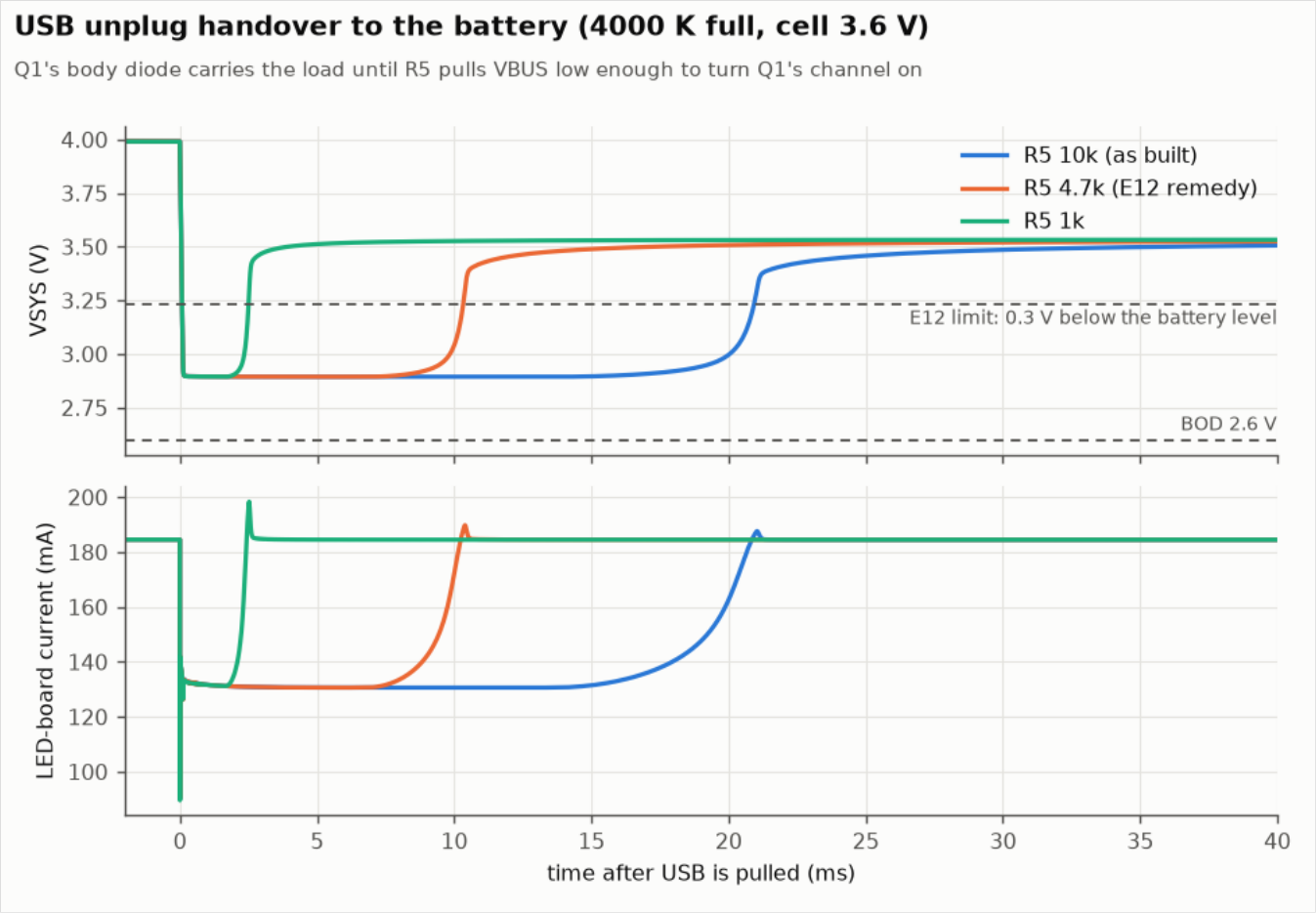
With a 0.1 Ω cable, VBUS peaks at 5.77 V (MCP73831 abs max 7 V; the TVS does not conduct) and VSYS at 4.78 V (ATtiny and LP5012 abs max 6 V). With 0.3 Ω the peaks are 5.06 V / 4.20 V. The charger supplies a 10 A peak for a few microseconds; USB 2.0 asks a device to present no more than 10 µF at attach, and C1 is 10 µF nominal, but D3 adds C3/C4 while VSYS_RAW sits below VBUS − 0.85 V. Most chargers ride through it; no action proposed.



5. Unplugging USB (handover to the battery)

On USB, VSYS comes through D3 and Q1 is held off because its gate sits on VBUS. When USB is pulled, the lamp first drains C1 through D3, then Q1's body diode takes the load (cell → VSYS_RAW, about 0.65 V drop). Q1's channel turns on only when R5 has discharged VBUS below VSYS_RAW − |VGS(th)|. Cell 3.6 V, lamp 4000 K full (the E12 condition).

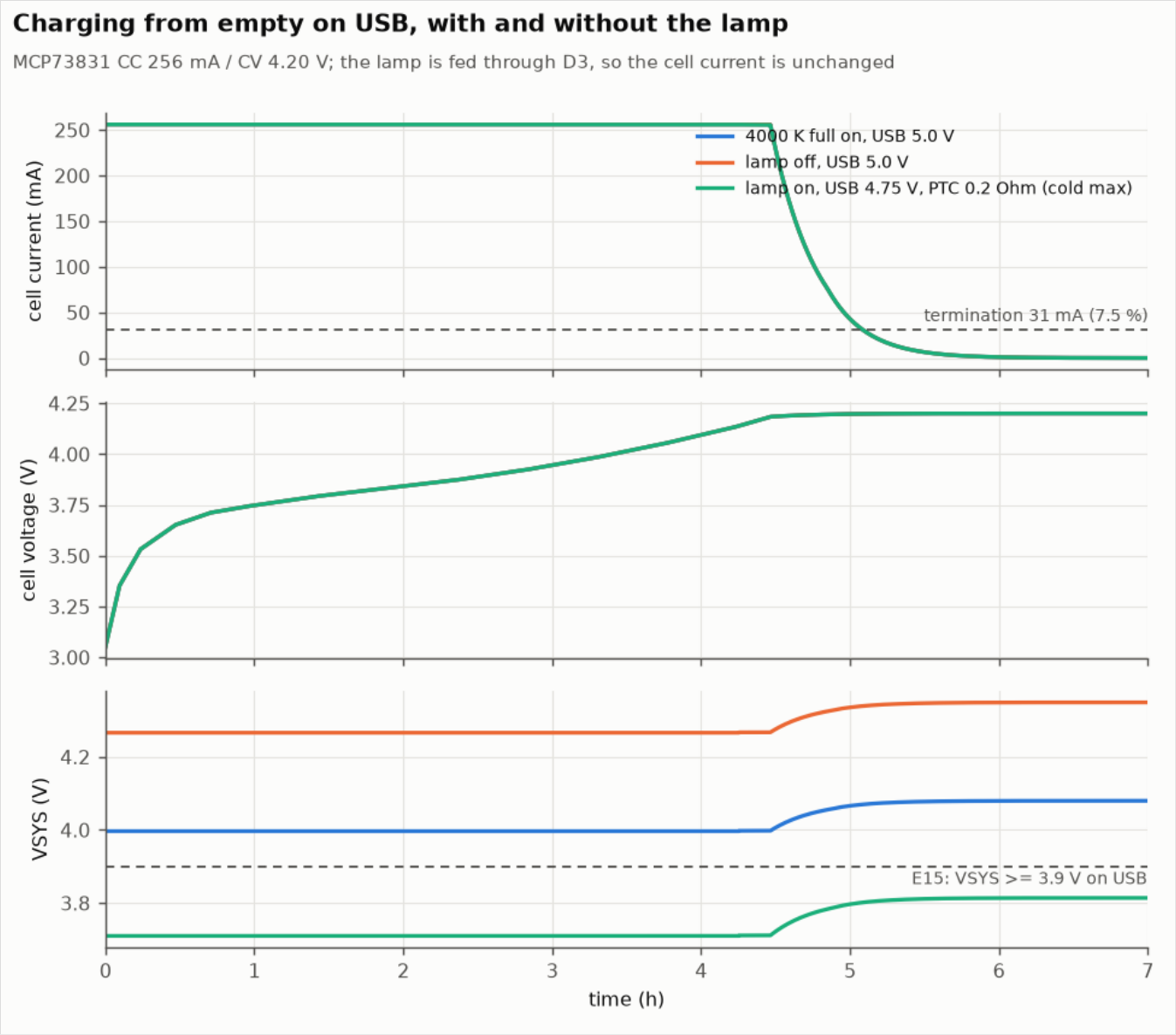
VSYS falls from 3.99 V to 2.89 V, 0.64 V below the 3.53 V it settles at on the battery, for 21 ms with R5 10k. R5 4.7k and 1k shorten this to 10 ms and 2.5 ms but do not reduce the depth. The MCU stays above its BOD; the whites lose headroom and the LED-board current drops by about 30 % for that time.



6. Charging with the lamp on (load sharing)

A full charge from an empty cell (3.0 V) on a 5.0 V source, with the lamp at 4000 K full for the whole charge, with it off, and on a weak source (4.75 V, PTC at its 0.2 Ω maximum); all through a 0.2 Ω USB cable. The MCP73831 model is CC 256 mA / CV 4.20 V; the 31 mA termination point is read off the cell current.

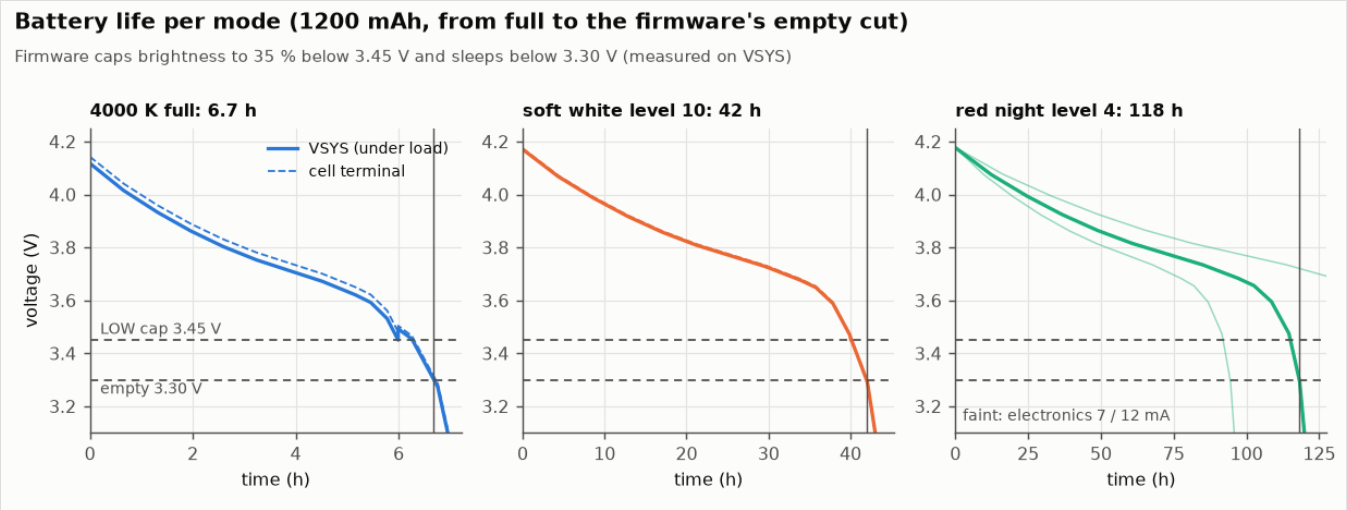
The cell current is the same with and without the lamp: the lamp is fed from D3, so the charger sees only the cell and reaches its 31 mA termination after 5.1 h either way (R18). Peak USB current 0.45 A (lamp on) and 0.26 A (off); charger dissipation peaks at 0.46 W. VSYS stays at 4.00 V or more on the 5.0 V source but drops to 3.71 V on the weak one.



7. Battery life per mode

Discharge from full at a fixed mode until VSYS (which the firmware measures under load) falls to 3.30 V. Below 3.45 V the firmware caps brightness to 35 %, modelled as a latch. LED drive is the average PWM duty on the LP5012 sinks; electronics are MCU 2.5 mA + LP5012 6.8 mA + MCP9800 0.2 mA. Thermal derating is not modelled, so the 4000 K figure is an upper bound once the head warms.

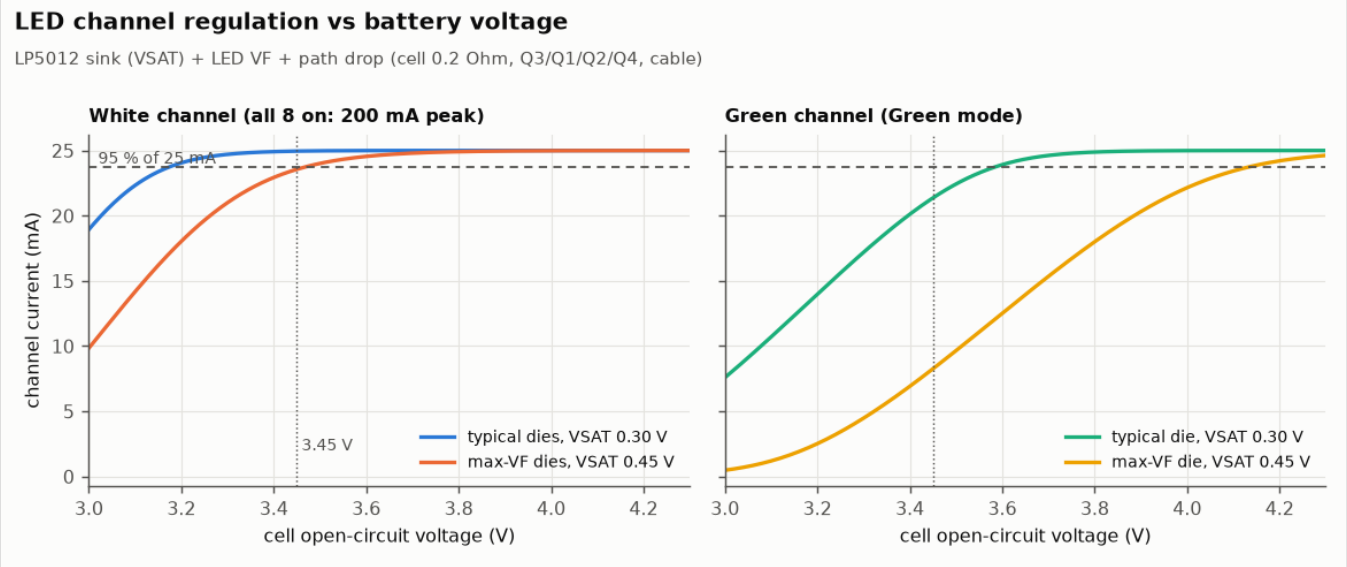
4000 K full: 6.7 h (brightness cap after 6.0 h; estimate 6.5 h). Soft white level 10: 42 h (estimate 45 h). Red night level 4: 118 h (estimate 100-150 h; 94-158 h for 12-7 mA of electronics, which dominate at this level). Every mode delivers about 4.45 Wh before the cut.



8. LED channel headroom

DC sweep of the cell's open-circuit voltage with 0.2 Ω cell resistance and the full path (Q3, Q1, Q2, Q4, cable). Left: all eight whites on at once, the instantaneous peak of 4000 K full (200 mA). Right: the Green mode. Typical dies with VSAT 0.30 V, and maximum-VF dies with VSAT 0.45 V (VSAT at 25 mA is not specified).

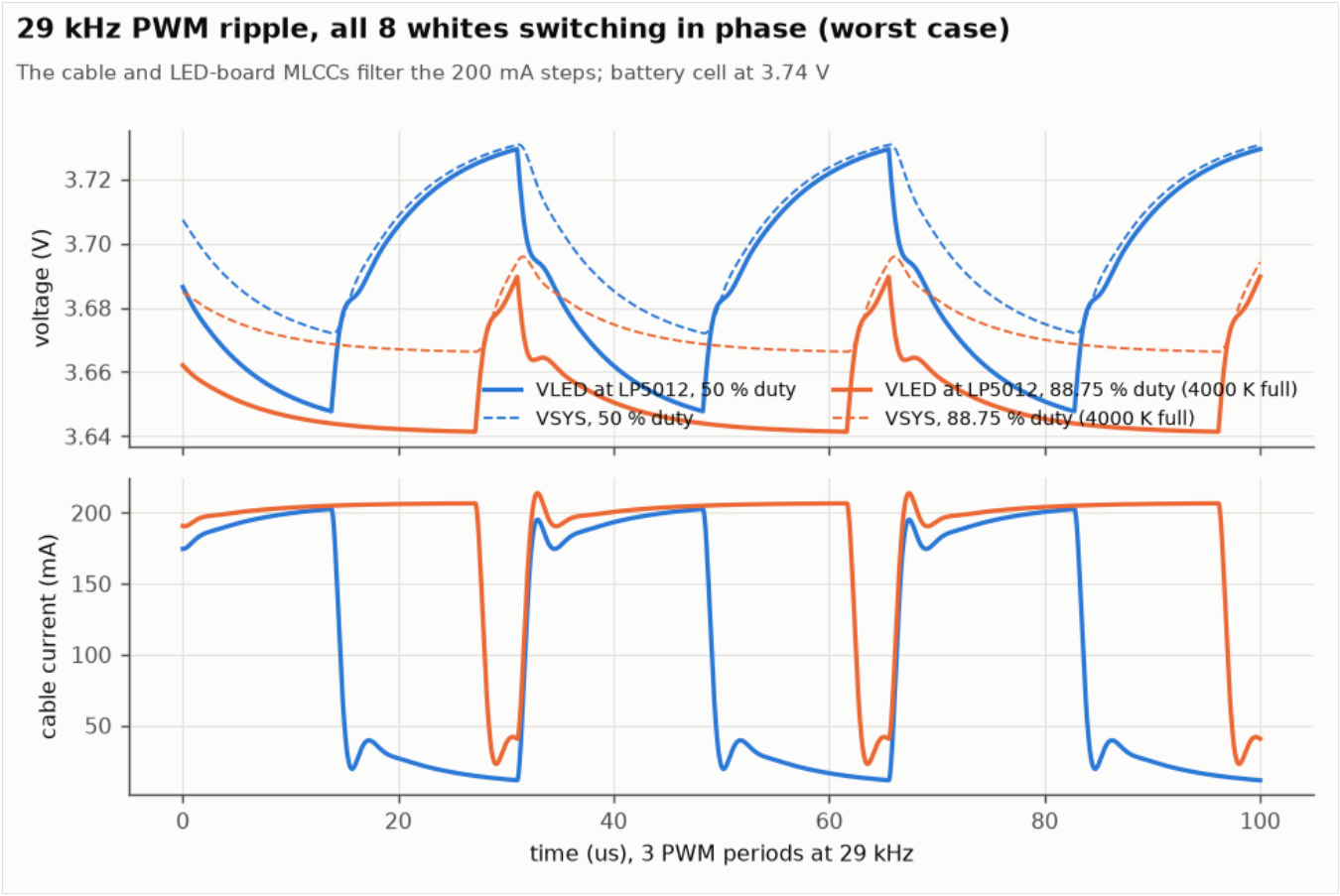
The whites hold 95 % of 25 mA down to VSYS 3.10 V (typical) and 3.40 V (worst case), so they are regulated wherever the firmware runs at full brightness. Green needs VLED 3.57 V at the board with a typical die (design note: ~3.65 V) and 4.12 V with a maximum-VF die (design note: 4.05-4.15 V). A worst-case die reaches only 15 mA at a 3.7 V cell. VERIFY E9 measures the received dies.



9. PWM ripple at 29 kHz

All eight whites switching together at the LP5012's 29 kHz, which is the worst case: the datasheet extract does not say whether the LP5012 staggers its outputs. 50 % duty gives the largest ripple; 88.75 % is 4000 K full.

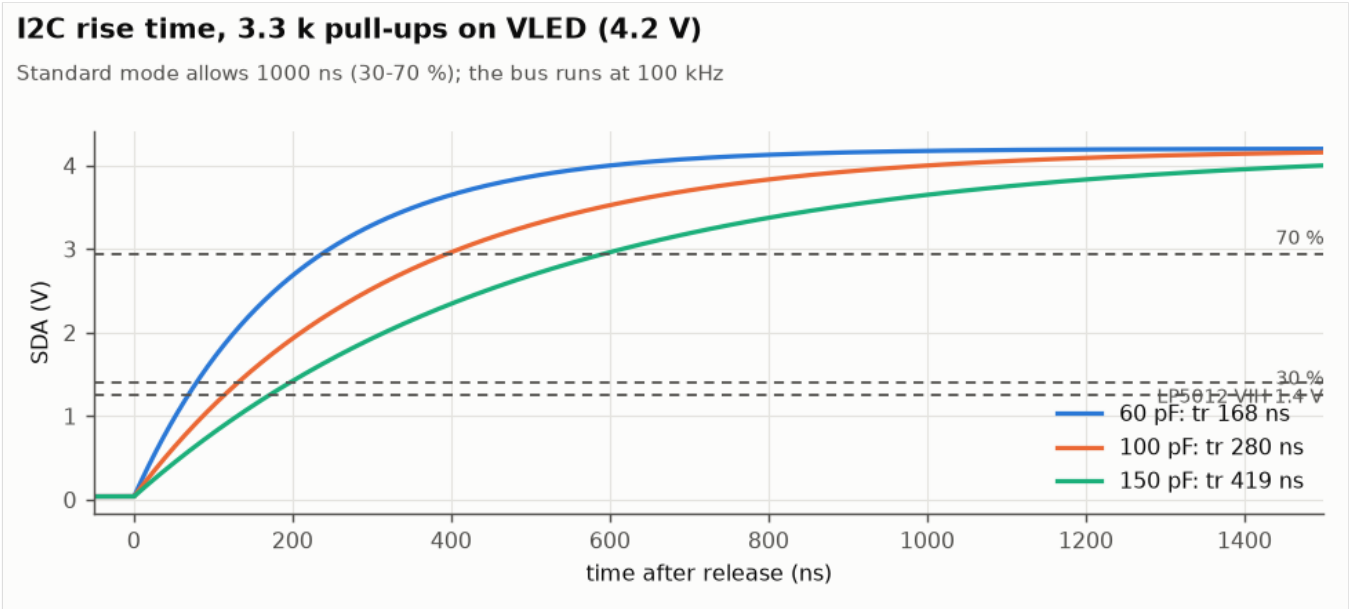
At 50 % duty, VLED at the LP5012 swings 82 mV p-p (minimum 3.65 V) and VSYS 59 mV p-p. Far from the LP5012's 2.7 V minimum and the MCU's BOD; the ATtiny's ADC reading of VDD will see this ripple, which the firmware averages.

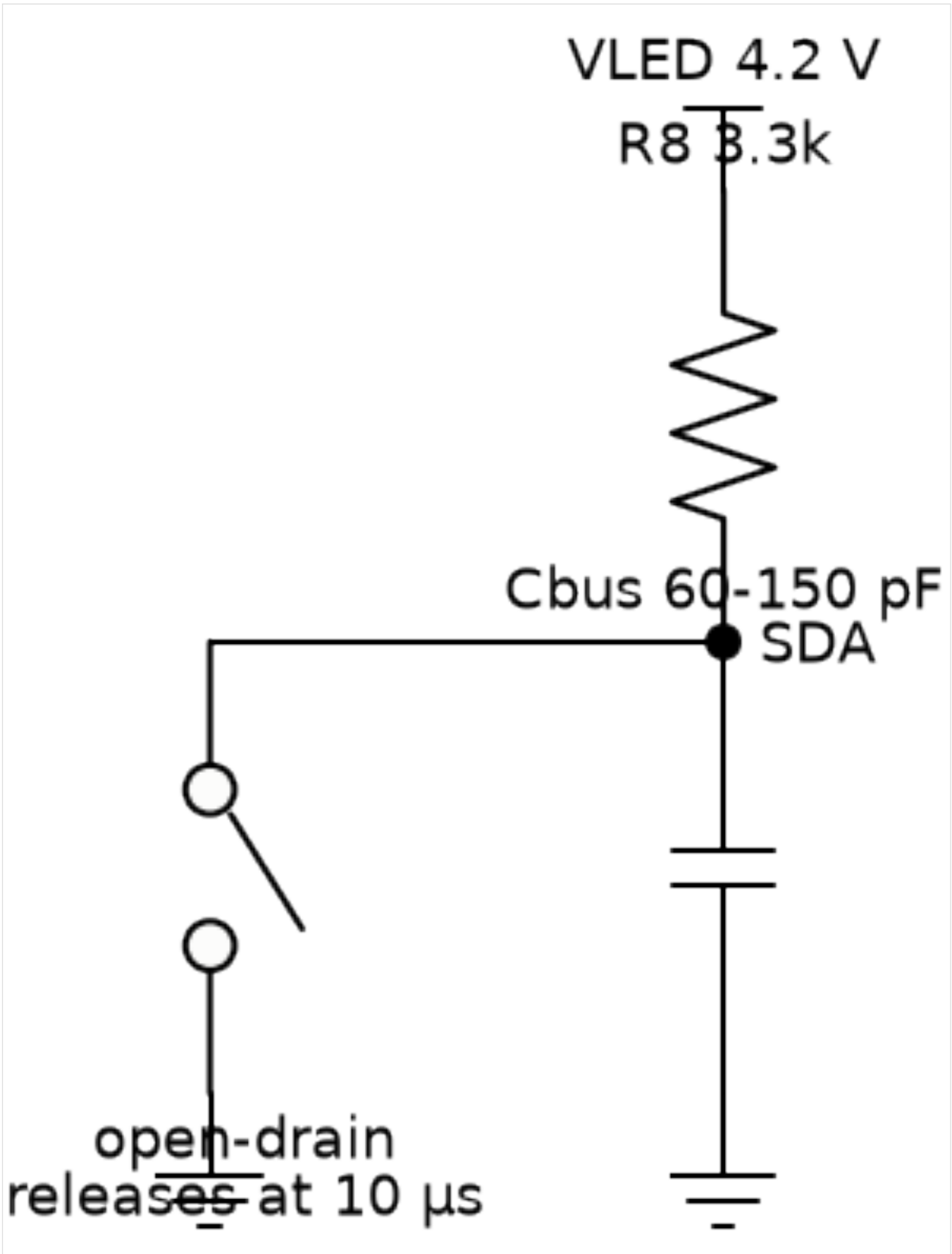


10. I²C rise time

R8/R9 3.3 kΩ to VLED against the bus capacitance (MCU pin, 200 mm cable, LP5012, MCP9800): 60-100 pF estimated, 150 pF as margin.

Rise time (30-70 %): 168 / 280 / 419 ns for 60 / 100 / 150 pF, against 1000 ns for standard mode, where the firmware runs the bus (100 kHz). Fast mode (400 kHz, 300 ns) would pass at 60 pF only, so the bus should stay at 100 kHz.





I²C test circuit: an open-drain pin releases SDA at 10 µs.

Model check

Each device model at the operating points it was fitted to (`sim/cases/model_check.cir`).

QUANTITY	MODEL	DATASHEET / DESIGN	SOURCE
DMG2305UX R_DS(on) @ VGS -4.5 V (Ohm)	0.040	0.040 typ / 0.052 max	KEYSPECS
DMG2305UX R_DS(on) @ VGS -2.5 V (Ohm)	0.084	0.052 typ / 0.100 max	KEYSPECS
DMG2305UX body diode @ 0.1 A (V)	0.653	~0.65 (PROVISIONAL)	typical silicon
Thrive 2835 white VF @ 10 mA (V)	2.701	~2.70	DS312 Fig. 9
Thrive 2835 white VF @ 25 mA (V)	2.775	~2.75	DS312 Fig. 9
Thrive 2835 white VF @ 50 mA (V)	2.851	~2.85	DS312 Fig. 9
Thrive white max-VF die @ 25 mA (V)	2.925	~2.92	electronics.md 2
Wurth red VF @ 20 mA (V)	2.001	2.0 typ	Wurth p2
Wurth red max die @ 20 mA (V)	2.401	2.4 max	Wurth p2
Wurth green/blue VF @ 20 mA (V)	3.201	3.2 typ	Wurth p2
Wurth green/blue max die @ 20 mA (V)	3.602	3.6 max	Wurth p2
Wurth G/B VF rise 20 -> 25 mA (V)	0.090	~+0.09	Wurth p3
S1J VF @ 0.2 A (V)	0.855	~0.85	PLAN / Vishay curve
S1J VF @ 1 A (V)	0.954	~0.95 typ (1.1 max)	Vishay
LP5012 sink current at VSAT (fraction of ISET)	0.950	0.95 (definition of VSAT)	SLVSEH2B

Assumptions (PROVISIONAL until measured)

VALUE	USED	REPLACED BY
Cell open-circuit voltage vs charge	generic LiCoO2 pouch curve (3.0-4.18 V)	a discharge log of the received cell (bench supply as a load, or F4's run)
Cell + PCM resistance	0.20 Ω; lead 150 nH	the cell's voltage step at a known load
Cell PCM over-current trip	~3 A (DW01-class)	the cell supplier's data
10 µF 0805 16 V DC bias	6.3 µF at 4.2 V	VERIFY E5
1 µF 0402 / 0603 DC bias	0.56 / 0.72 µF at 4.2 V	manufacturer curves
DMG2305UX Crss and body diode	100 pF; 0.65 V at 0.1 A	datasheet curves
LP5012 VSAT at 25 mA	0.30 V typ, 0.45 V max	VERIFY E9
LED forward voltage bins	typical and maximum dies	VERIFY E9 (received dies)
LP5012 + MCP9800 supply	6.8 mA active, 10 µA standby	VERIFY E7
MCU supply	2.5 mA	VERIFY E3
Neck cable	63 mΩ and 75 nH per conductor	a 4-wire resistance measurement
USB cable	0.1-0.3 Ω, 1 µH	the charger and cable in use
VSYS bulk option	47 µF, 0.1 Ω ESR	the part chosen, if any

Not simulated

- Thermal behaviour: the firmware's 45-58 °C derating, the MCP9800 thermostat and F1's trip are thermal (seconds to minutes) and need the T-series bench tests.

- The MCP73831's control loop, thermal regulation, precharge timer and termination logic (behavioural CC/CV only).
- The ATtiny's BOD response time and the LP5012's start-up sequence (UVLO only).
- ESD on USB-C and the buttons, and the TVS's dynamic resistance beyond its datasheet clamp point.
- Whether the LP5012 staggers its PWM edges (the ripple case assumes it does not).
- Layout parasitics: copper resistance and inductance on both boards are not included.

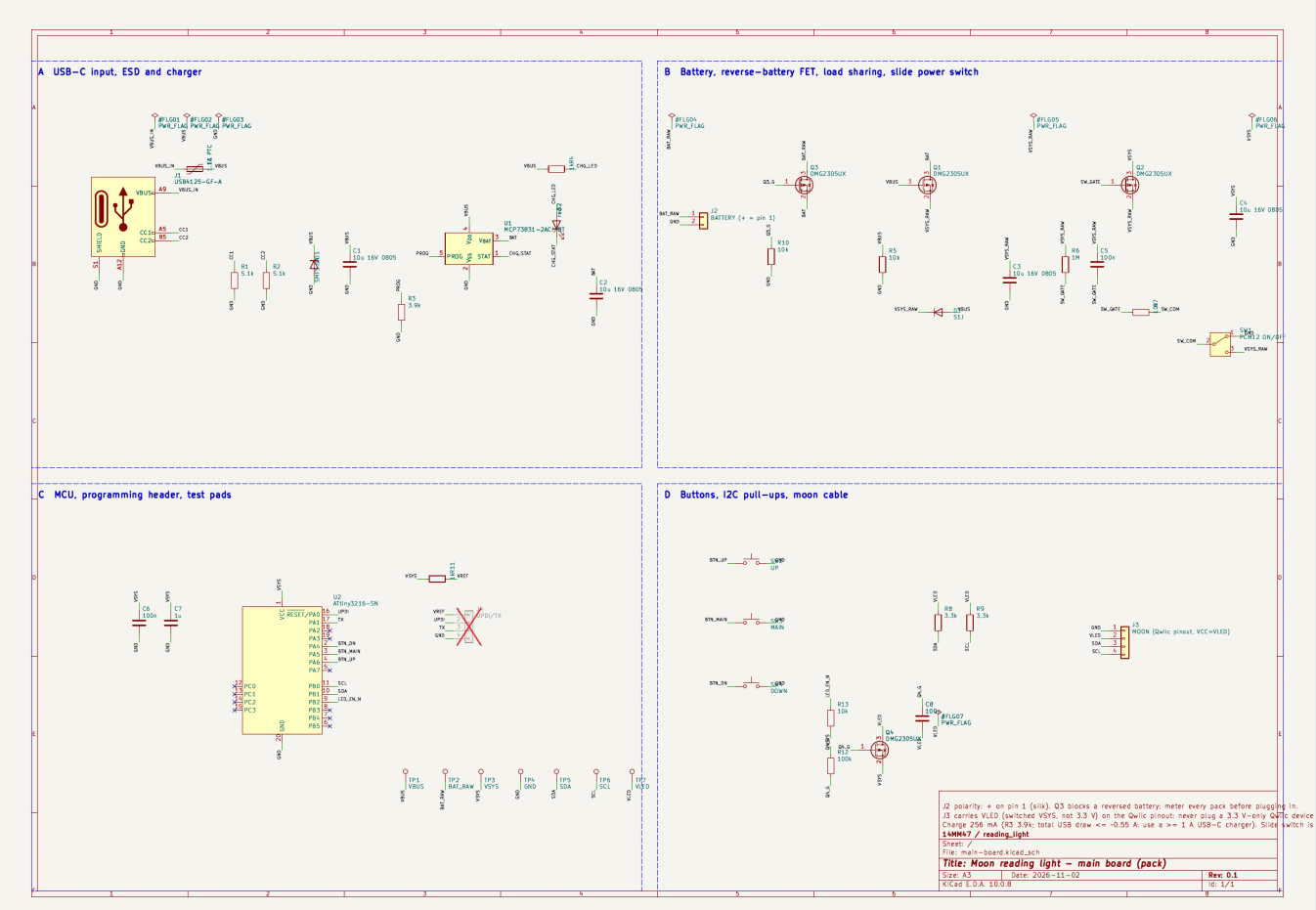
Reproduce

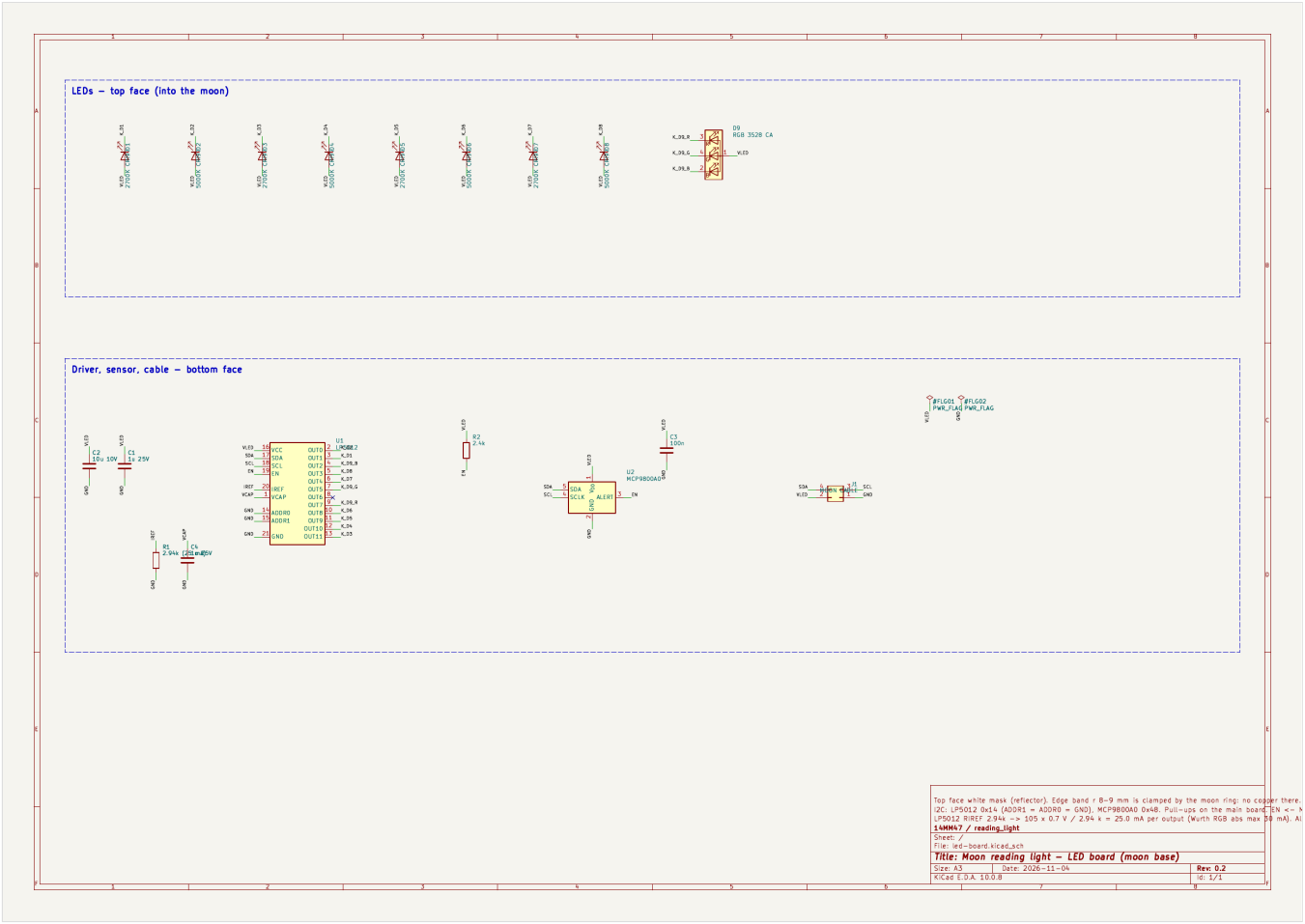
```
python3 sim/run_sims.py      # ngspice 42, numpy, matplotlib, schemdraw; ~30 s
```

Netlists as run, ngspice logs and raw waveforms land in `sim/out/<case>/<variant>/`. To test a design change, edit `sim/boards.inc` (or add a variant in `sim/run_sims.py`) and re-run; replace PROVISIONAL values in `sim/models.inc` as bench measurements come in.

Appendix: board schematics (reference)

Exported from the generated KiCad schematics for comparison with the simulated circuits above.




LEDs – top face (into the moon)
D9
RGB 3528 CA
Driver, sensor, cable – bottom face
U1
LP5012
U2
MCP9800A0
R2
2.4k
C3
100n
C2
10u 10V
C1
1µ 25V
R5
2.94k
Ck
220n 25V
#FLG01
PWR_FLAG
#FLG02
PWR_FLAG
Top face white mask (reflector). Edge band r 8–9 mm is clamped by the moon ring; no copper there.
I2C: LP5012 0x14 (ADDR1 = ADDR0 = GND), MCP9800A0 0x48. Pull-ups on the main board. EN <- N
LP5012 RIREF 2.94k -> 105 x 0.7 V / 2.94 k = 25.0 mA per output (Wurth RGB abs max 30 mA). Al
1AMMA7 / reading_light
Sheet: /
File: led-board.kicad_sch
Title: Moon reading light – LED board (moon base)
Size: A3
Date: 2026-11-04
Rev: 0.2
KiCad E.D.A. 10.0.8
Id: 1/1